

David Bian

Software Engineer | AI & Backend | Open to Relocation

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EDUCATION

Duke University

Bachelor of Science in Computer Science and Biology

Aug. 2022 – May 2026

3.8/4.0 GPA

PROFESSIONAL EXPERIENCE

Research Software Engineer

January 2025 – May 2026

Duke Center for Virtual Imaging Trials

- **Equipped 8 researchers** to run cross-platform experiments by building a Python pipeline with REST APIs designed to eliminate manual configuration.
- **Saved 5+ hours** of compute/debugging per failed run by adding a JSON Schema validation service to catch malformed and invalid inputs.
- **Enabled 7 new features** to be added without modifying existing modules by refactoring a Python research workflow into loosely coupled components.

Machine Learning Research Intern

May 2024 - July 2024

Duke University

- Built a containerized diffusion model pipeline (PyTorch, Hugging Face Diffusers, Docker) to generate class-conditional images for any CIFAR-10 minority class on demand.
- **Automated 124+ GPU experiments** in a Slurm-managed HPC environment to extensively benchmark diffusion-based augmentation against SMOTE.
- **Cut wasted GPU hours by ~17%**, by automating visual regression checks across 30+ model checkpoints, filtering low-quality outputs.

Software Design Teaching Assistant

January 2024 - May 2024

Duke University

- Audited 6 Java project teams' codebases; Identified SOLID violations, code smells, and design pattern misuse, with recommendations to improve code modularity and maintainability.
- Trained 6 developers in Agile, TDD, and Git best practices through 1:1s, reducing integration conflicts.

Algorithms Teaching Assistant

January 2024 - May 2026

Duke University

- Debugged 300 code submissions across Python, Java, and C++; traced root causes in algorithm design, edge-case handling, and asymptotic performance.

PROJECTS

Distributed LLM Evaluator

June 2026 - Present

Python, LangChain, Chroma, ONNX, Flask, Docker

- Built a containerized RAG agent framework on schema-validated outputs and tool-use retrievals.
- **Processed 9,000 API calls** in an hour by redesigning a sequential LLM pipeline for concurrency (asyncio).
- **Ensured reliable 150-replicate execution** by implementing a distributed system worker pool (SQLite-backed atomic claim queue).

Chesseval

React, TypeScript, Stockfish (WASM), WebSocket, FastAPI, SQL

- Built a client-side-first chess platform with live Stockfish evaluation running Stockfish in a Web Worker for offline analysis and using **FastAPI/WebSockets** for real-time multiplayer gameplay.

SKILLS

Languages: Python, Java, C/C++, SQL, bash, MATLAB, JavaScript, TypeScript

Tools: Docker, Chroma, React, Linux, Git, GitHub Actions, CI/CD, Slurm, AWS, GCP

Libraries: PyTorch, Diffusers, FastAPI, LangChain